



Network Technologies

Introduction - Case Study

When the project is finished, the outcome will be an IPv4 addressing scheme that has been planned and constructed according to requirements that are appropriate for the network architecture that was described earlier. This scheme will be the result of the project's completion. The private network addresses will be of the Class C variety, with the pattern 192.168.xyz.0/24, and the subnet mask will either be 255.255.255.0 or /24. The private network will not be accessible to the public. These will serve as the foundation upon which the design is built. You will have a simpler time choosing the appropriate values for x, y, and z if you follow these guidelines and do it in the following order: If your birthday falls in the months of January, February, March, or April, you should substitute 0 for the value of x in the formula. If, on the other hand, your birthday falls in the months of July or August, you should substitute 1 for x in the calculation since July and August are months that fall in September and December, respectively. In addition to this, the y and z digits of the final two numbers on your student ID will need to be modified in order to reflect the new information. Given that my student identification number is 30090498 and that my birthday is July 18, 1997 (x = 1), the number yz = 98 is a number that accurately describes me. This is because my student identification number is a four-digit numeric value. As a consequence of this, I will be putting the IP address range that was mentioned before to use for the activity that was mentioned. 192.168.198.0/24.

Sub netting

It is necessary for us to divide the Network Address Space into four separate subnets. The first three subnets will be utilized in order to make room for the three local area networks. The three Point-to-Point Wide Area Networks will be derived from the fourth subnet when it is subnetted.

LAN	Users	IP Network	Subnet	Broadcast address	Usable IP Range
LAN 1	60	192.168.198.0	255.255.255.192	192.168.198.63	192.168.198.1 – 192.168.198.62
LAN 2	58	192.168.198.64	255.255.255.192	192.168.198.127	192.168.198.65 – 192.168.198.126.
LAN 3	48	192.168.198.128	255.255.255.192	192.168.198.191	192.168.198.129

					– 192.168.168.190
R1-R2	2	192.168.198.252	255.255.255.252	192.168.198.255	192.168.198.253 – 192.168.168.254
R1-R3	2	192.168.198.248	255.255.255.252	192.168.198.251	192.168.198.249 – 192.168.168.250
R2-R3	2	192.168.198.244	255.255.255.252	192.168.198.247	192.168.198.245– 192.168.168.246

LAN 01

Gateway Address: 192.168.198.1

Users: 192.168.198.2 - 192.168.198.62

LAN 02

Gateway Address: 192.168.198.65

Users: 192.168.198.66 - 192.168.198.126

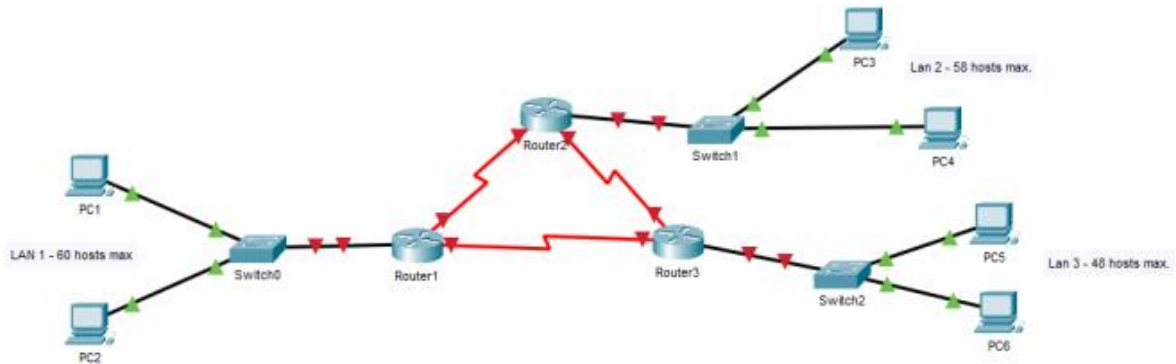
LAN 03

Gateway Address: 192.168.198.129

Users: 192.168.198.130 - 192.168.198.190

Implementation in Packet Tracer

Design Topology



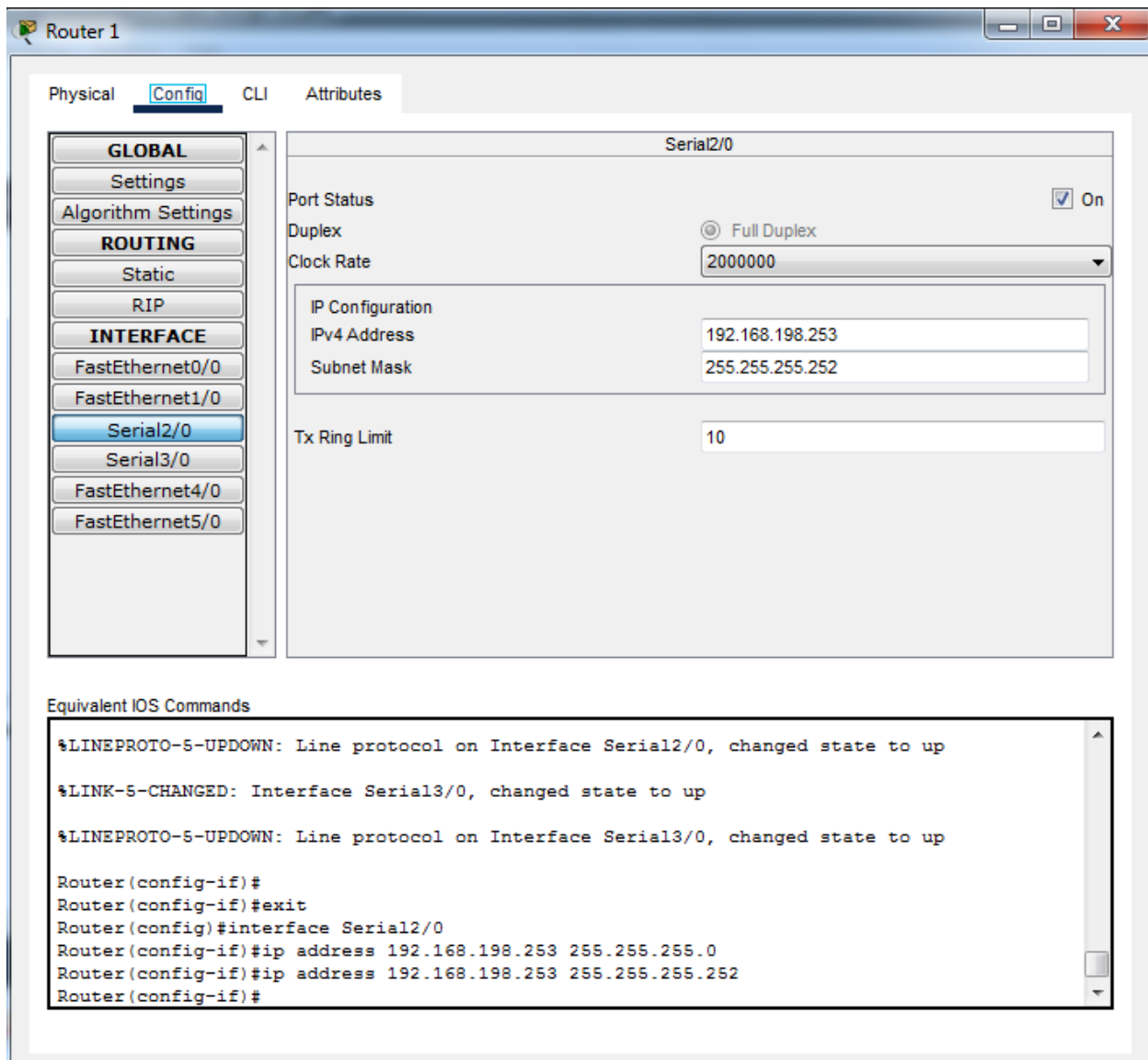
Configuration of WAN IP in Routers

Please activate your PC and initiate the execution of Packet Tracer. To initiate the execution of Packet Tracer, users can opt for one of two methods: either by performing a double-click action on the desktop shortcut associated with the software, or by manually going to the directory where the executable file of the program is stored and executing it from there. Upon initiating the Packet Tracer application, the user will be presented with an unoccupied workspace featuring a Logical topology, as seen in the accompanying visual representation.

Router 01

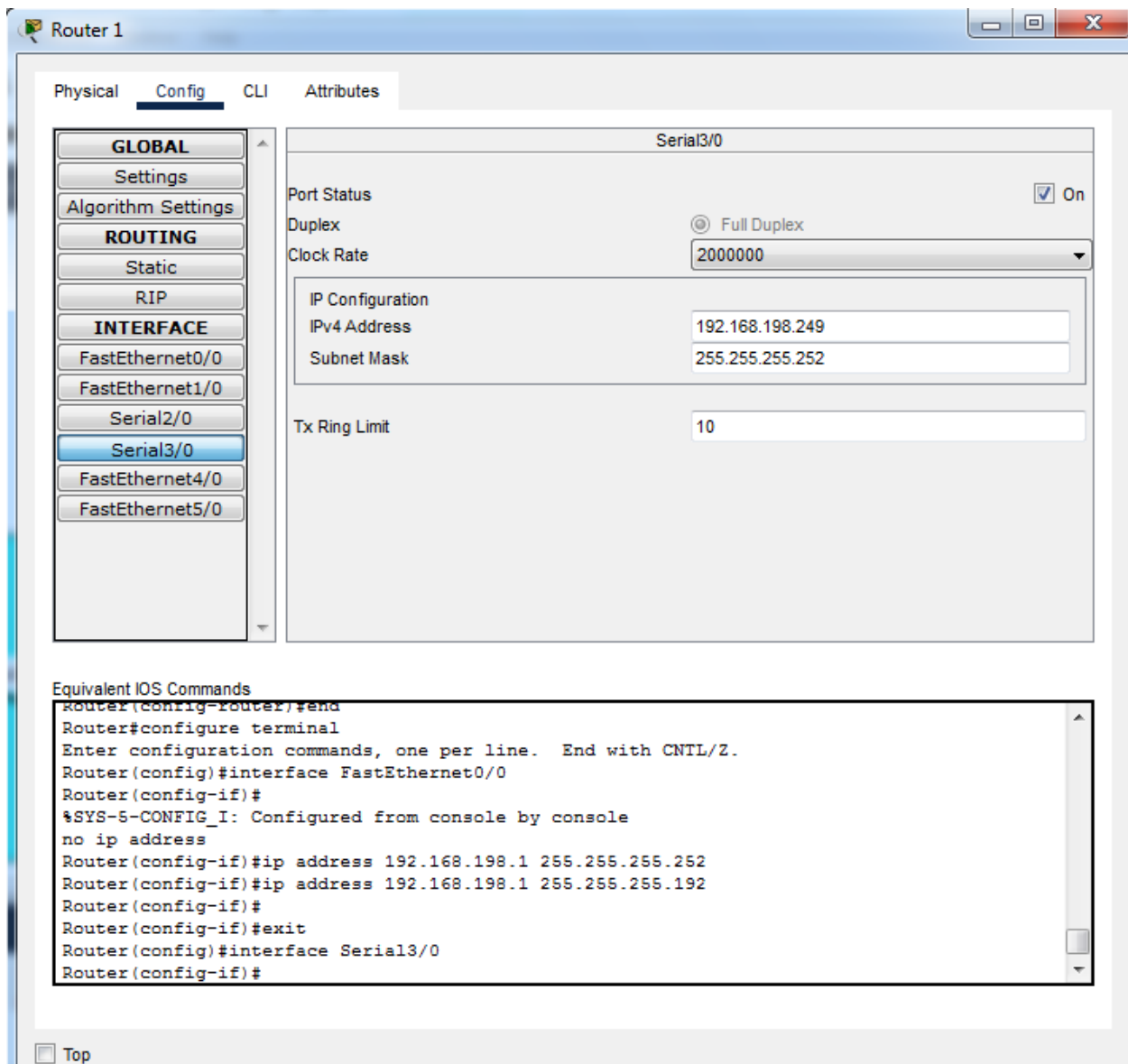
The command line interface, more often referred to as CLI, was provided on the very first router ever created. Due to the fact that this is a need, it is absolutely necessary that IP addresses be allocated to the interfaces of Router 1. In order for the router to fulfill its function as a data relay across networks, it is necessary to provide it with an Internet Protocol address (IP address).

The IP address 192.168.198.253 is being assigned to the serial 2/0 interface that is now being used. This step in the process of configuring the system is currently taking place.



The first routers used a command line interface, or CLI for short. Due to this necessity, assigning IP addresses to the interfaces of Router 1 is crucial. Assigning the router with an IP address makes it possible for it to act as a data relay across networks. The currently active serial 3/0

interface has the IP address 192.168.198.249. This part of the process of setting up the system is now happening.



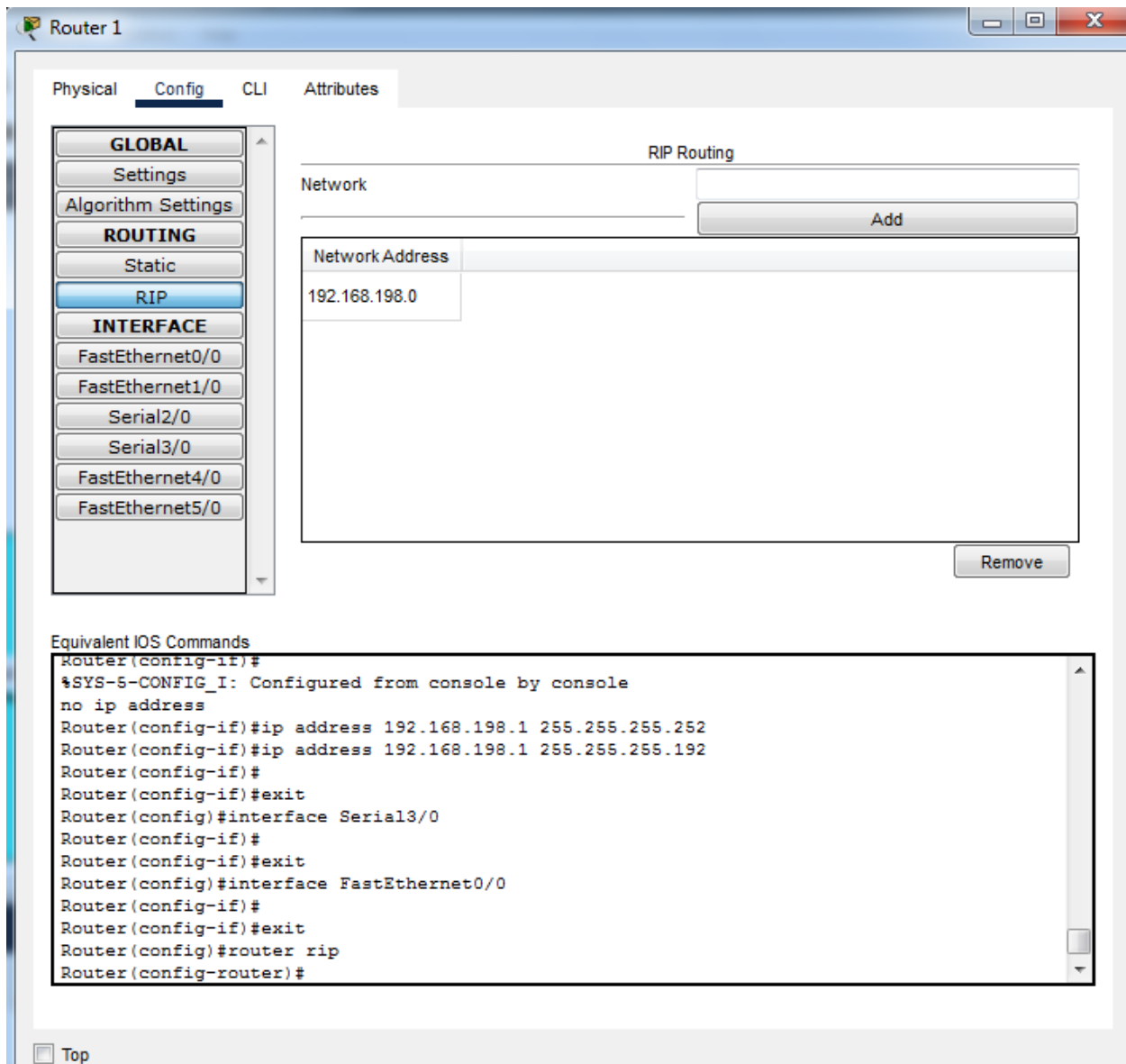
Similarly, we were tasked to use and design the three LANs that must be deployed and used in the network. The IP address that we have used in the network have the P address as 192.168.198.1 and the subnet that we have used is 255.255.255.0.

The screenshot shows the configuration window for Router 1, specifically the **FastEthernet0/0** interface. The interface is configured with the following settings:

- Port Status:** ☒ On
- Bandwidth:** ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex:** ☐ Half Duplex ☒ Full Duplex ☒ Auto
- MAC Address:** 0060.7074.60A5
- IP Configuration:**
 - IPv4 Address:** 192.168.198.1
 - Subnet Mask:** 255.255.255.192
- Tx Ring Limit:** 10

Below the configuration window, the **Equivalent IOS Commands** are listed:

```
Router(config)#interface FastEthernet0/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console
no ip address
Router(config-if)#ip address 192.168.198.1 255.255.255.252
Router(config-if)#ip address 192.168.198.1 255.255.255.192
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
```



Router 02

If you select router0, you may give it an IP address. To proceed, head to the settings page and select "Interfaces." The next step is to use the IP addressing Table to give each of the FastEthernet and serial ports an IP address. The section for IPv4 address and mask information.

Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 192.168.198.254

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config)#router rip
Router(config-router)#network 192.168.198.0
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
```

Router2

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status

☒ On

Duplex

Full Duplex

Clock Rate

2000000

IP Configuration

IPv4 Address

192.168.198.245

Subnet Mask

255.255.255.252

Tx Ring Limit

10

Equivalent IOS Commands

```
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#
```

Router2

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex☒ Full Duplex

☒ Auto

MAC Address0007.EC22.9122

IP Configuration

IPv4 Address

Subnet Mask

Tx Ring Limit10

Equivalent IOS Commands

Router(config)#interface FastEthernet0/0

Router(config-if)#

%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#

Router(config-if)#exit

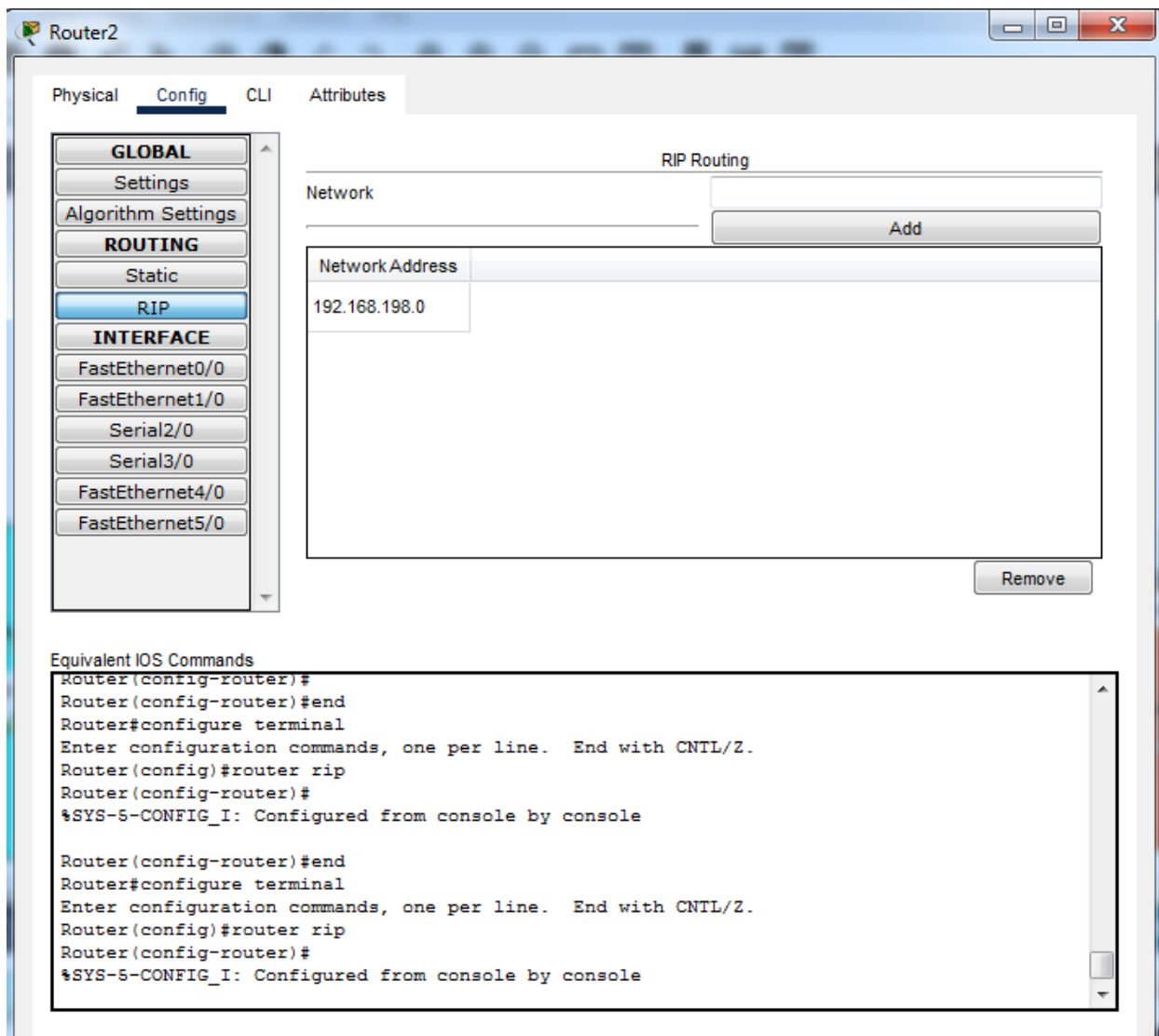
Router(config)#interface Serial3/0

Router(config-if)#

Router(config-if)#exit

Router(config)#interface FastEthernet0/0

Router(config-if)#



Router 03

Router3

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 192.168.198.250

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config)#router rip
Router(config-router)#network 192.168.198.0
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
```

Router3

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 192.168.198.246

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial3/0
Router(config-if)#
```

Router3

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060.4738.EC43

IP Configuration

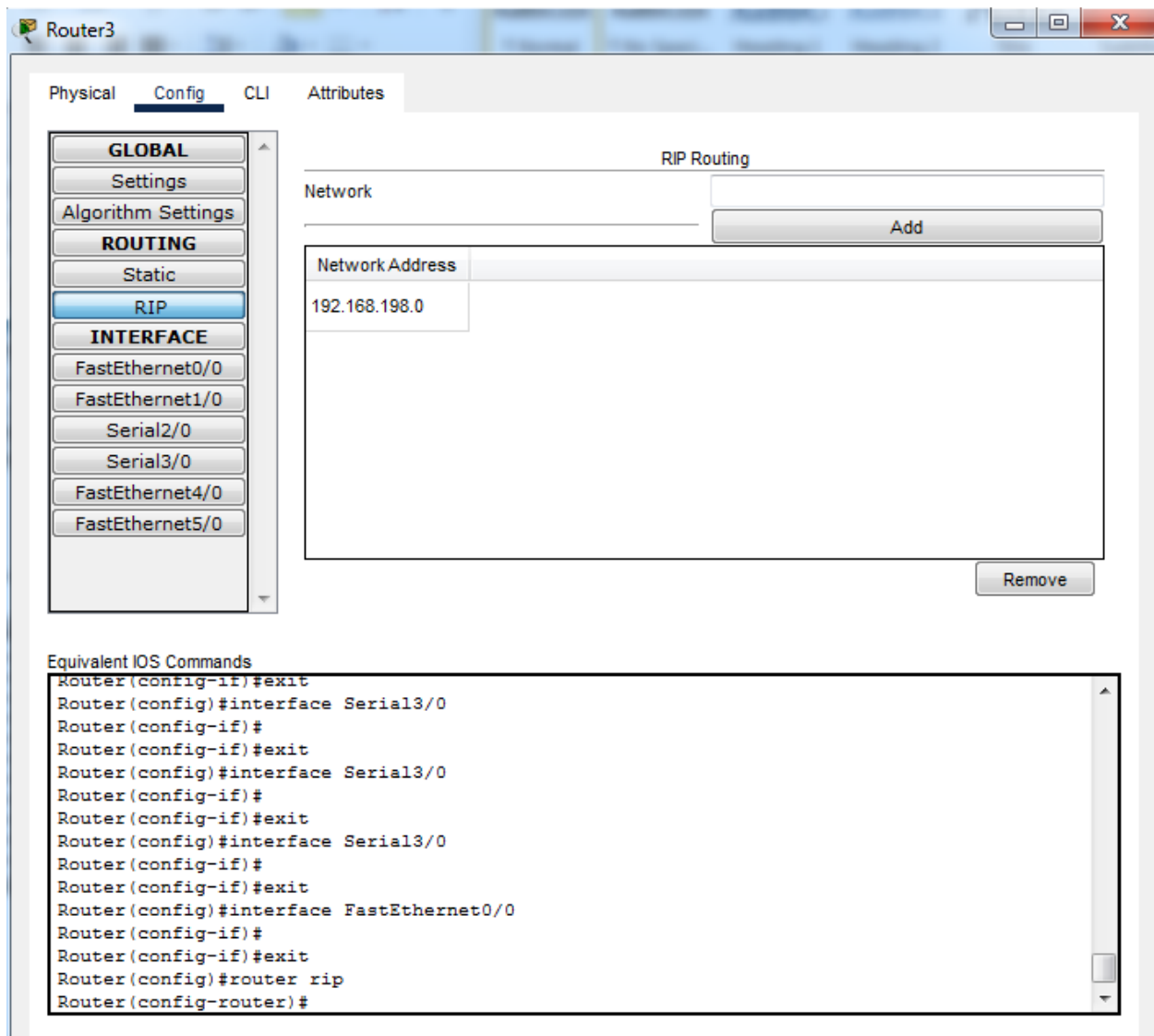
IPv4 Address

Subnet Mask

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial3/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial3/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial3/0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface FastEthernet0/0  
Router(config-if)#
```



Configuration of LAN 01 PCS

Please proceed by selecting PC0 so that we can get started with the procedure. Proceed to the prompt for the Disk Operating System (DOS). In order to complete the ping command, the IP address of the target node must be entered. The offered visual representation demonstrates the receipt of responses, so indicating that the connection is in a suitable state.

The PC0 possesses the capacity of being recognized as the recipient of an IP address, which may subsequently be allocated to it. This process can take place after the PC0 has been given the IP address.

After that, go to the desktop interface in order to acquire the IPv4 configuration, and once you're there, select the IP configuration option from the list of alternatives that are accessible to you.

It is necessary to fill in the IPv4 address, as well as the subnet mask and the default gateway, in the text boxes that are specifically designed for those pieces of information.

It is essential that all of the computers connected to LAN1 have the identical settings for their setup. The IP address is given to PC is 192.168.198.2 Subnet mask as 255.255.255.192.

PC0

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Bandwidth

☒ 100 Mbps

☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex

☒ Full Duplex

☒ Auto

MAC Address

0004.9A14.C0B4

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.198.2

Subnet Mask

255.255.255.192

IPv6 Configuration

☐ Automatic

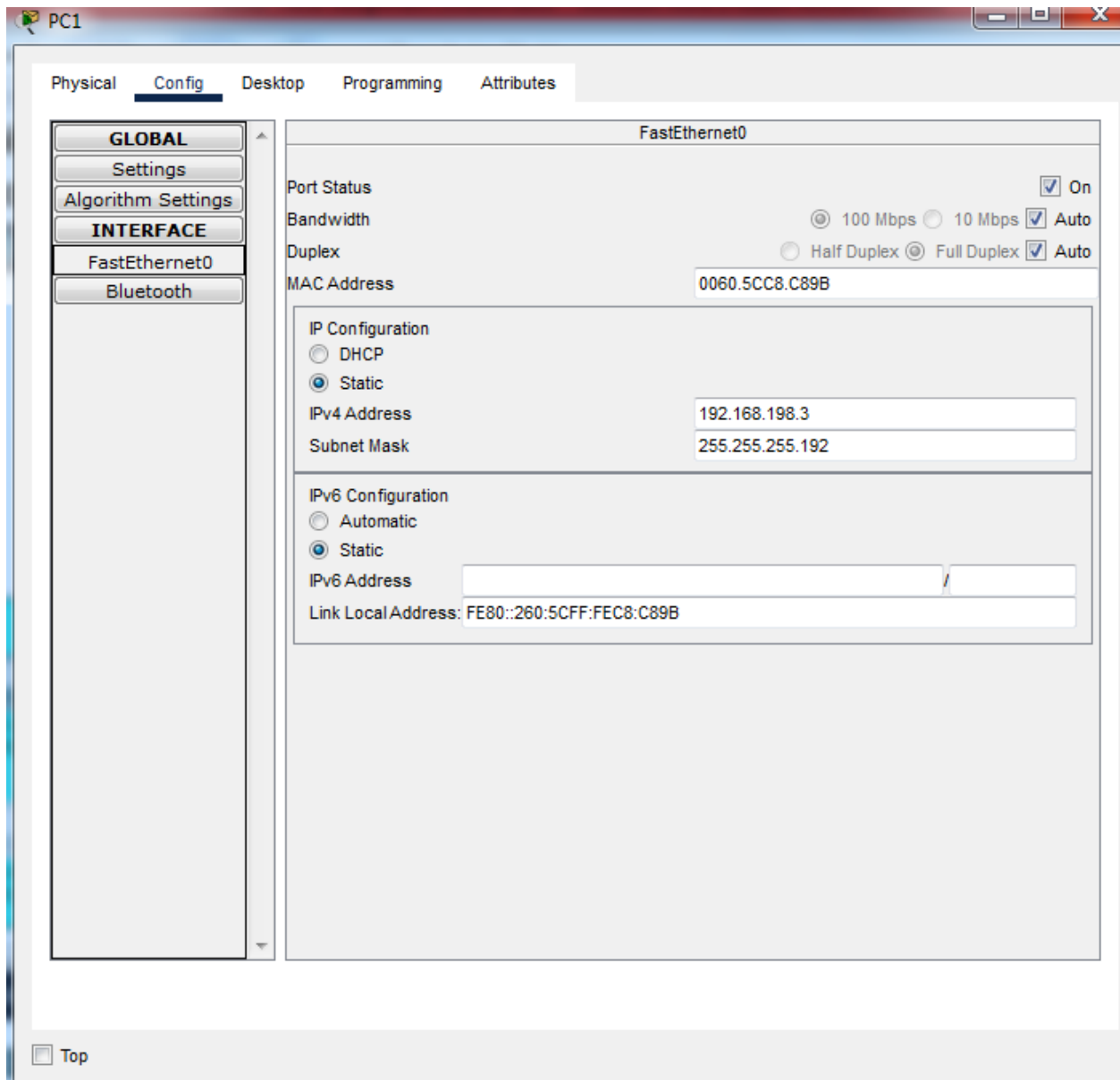
☒ Static

IPv6 Address

Link Local Address

FE80::204:9AFF:FE14:C0B4

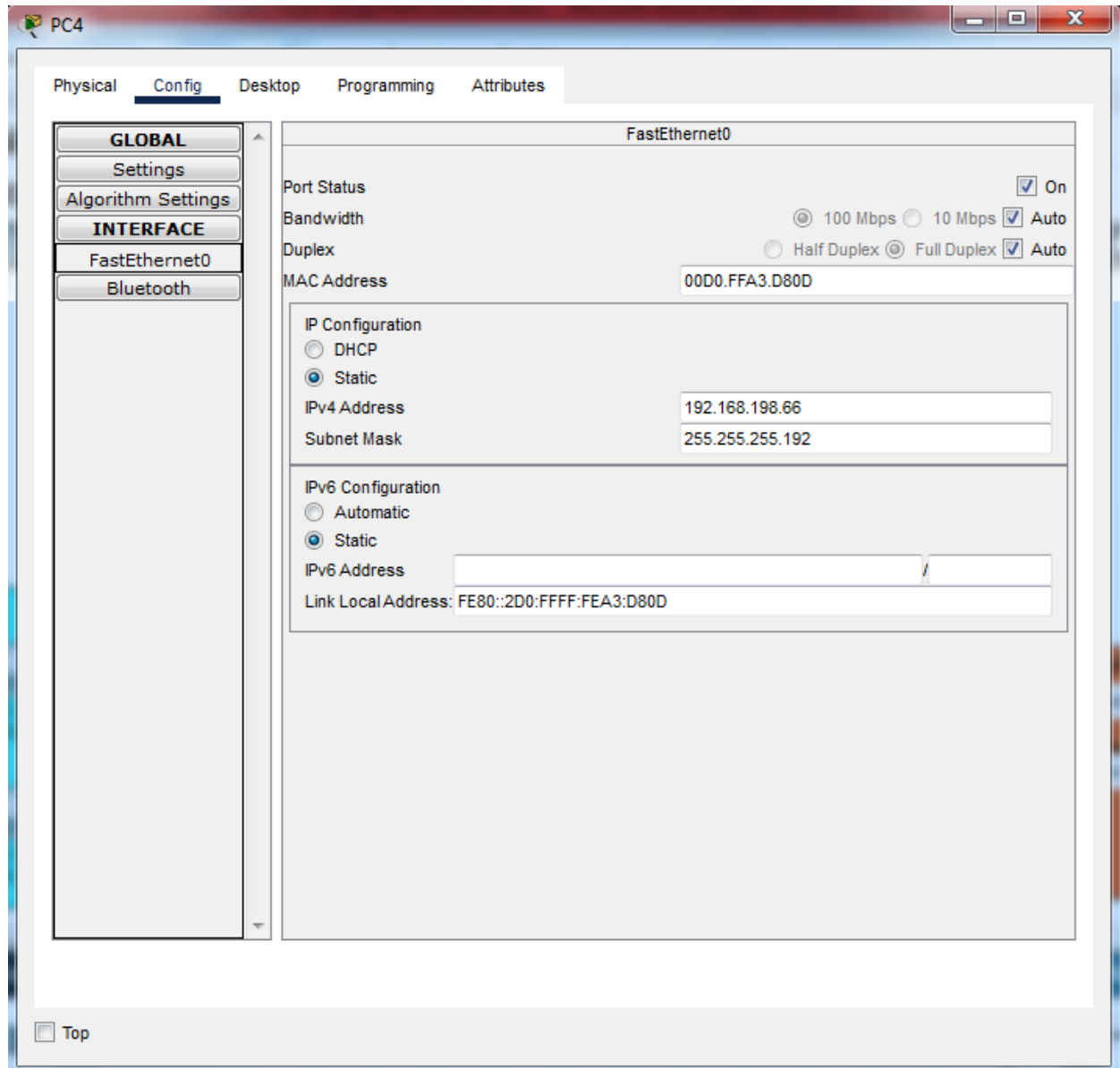
Top

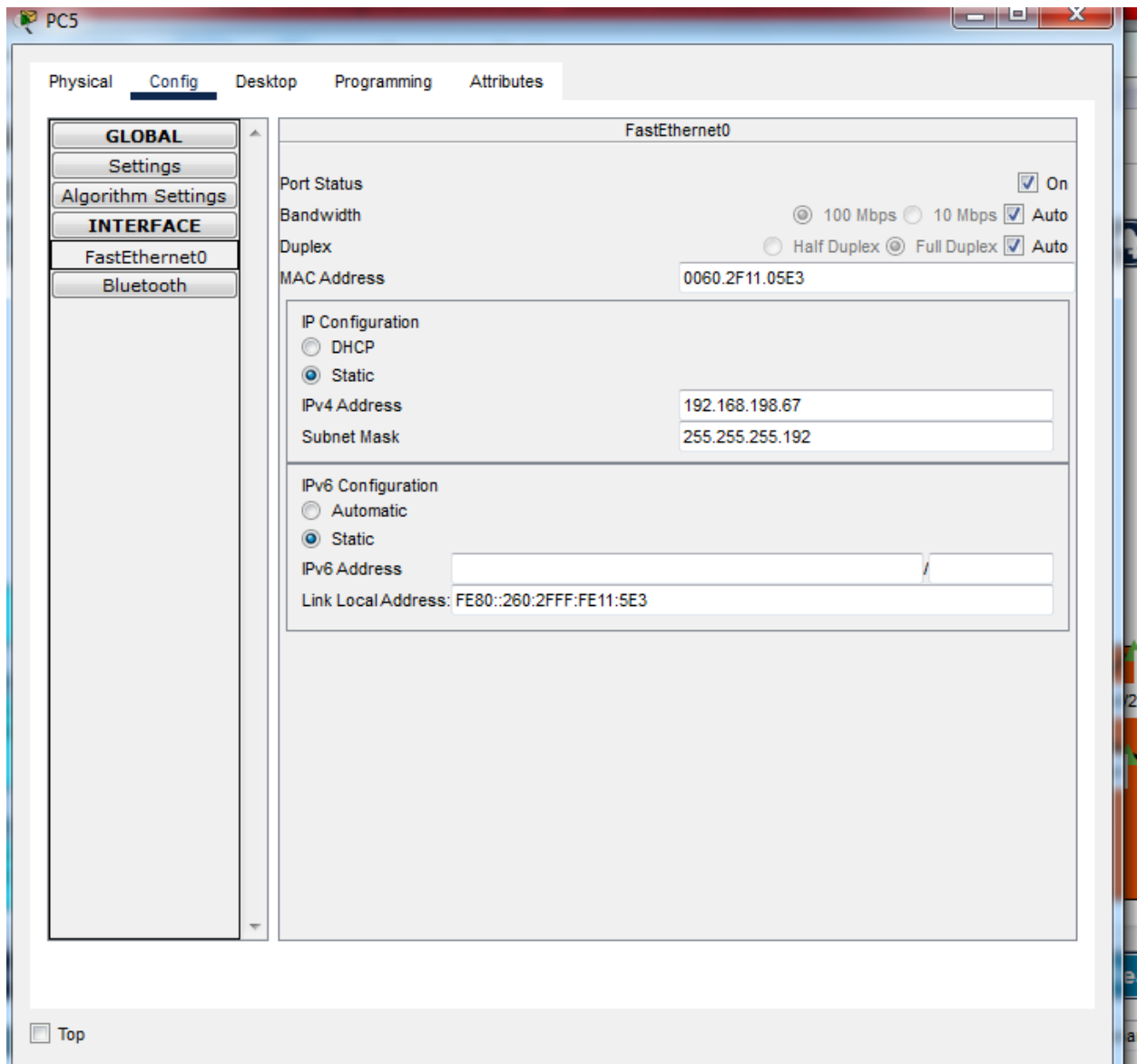


Configuration of LAN 2 PCs

The PC0 has the capability of being identified as the intended recipient of an IP address. After assigning an IP address to PC0, this step can be taken. The next step is to enter the desktop interface and choose the IP configuration option from the available possibilities to obtain the IPv4 configuration. Each component of the IPv4 address, including the subnet mask and the default gateway, must be entered into its respective text field. It is crucial that all LAN2 users'

PCs are configured in precisely the same way. The IP Address that I have used is 192.168.198.66 and subnet mask that is used 255.255.255.192.





Configuration of LAN 3 PCs

It is possible to identify the PC0 as the IP address's intended destination. This step can be completed after PC0 has been given an IP address. To acquire the IPv4 configuration, the next step is to open the desktop interface and choose the IP configuration option from the list of options. Every element of the IPv4 address, such as the default gateway and subnet mask, has to be typed into the corresponding text box. It is imperative that the PC configurations of every LAN2 user be exactly the same.

PC2

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Bandwidth

☒ 100 Mbps☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex☒ Full Duplex

☒ Auto

MAC Address00D0.BA4E.1287

IP Configuration

☐ DHCP

☒ Static

IPv4 Address192.168.198.130

Subnet Mask255.255.255.192

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::2D0:BAFF:FE4E:1287

☐ Top

PC3

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Bandwidth

☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address

0060.5CDC.2315

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.198.131

Subnet Mask

255.255.255.192

IPv6 Configuration

☐ Automatic

☒ Static

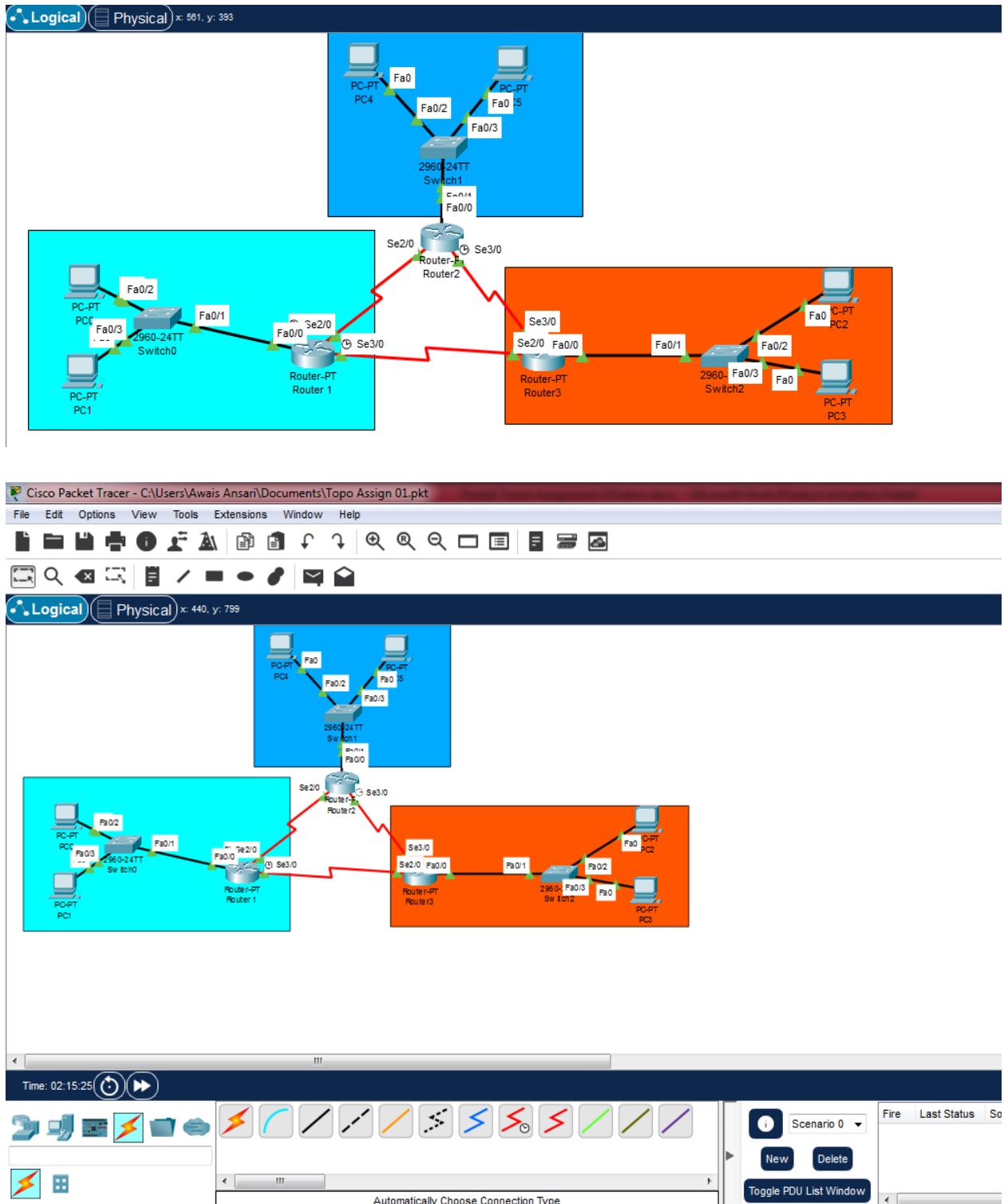
IPv6 Address

Link Local Address

FE80::260:5CFF:FEDC:2315

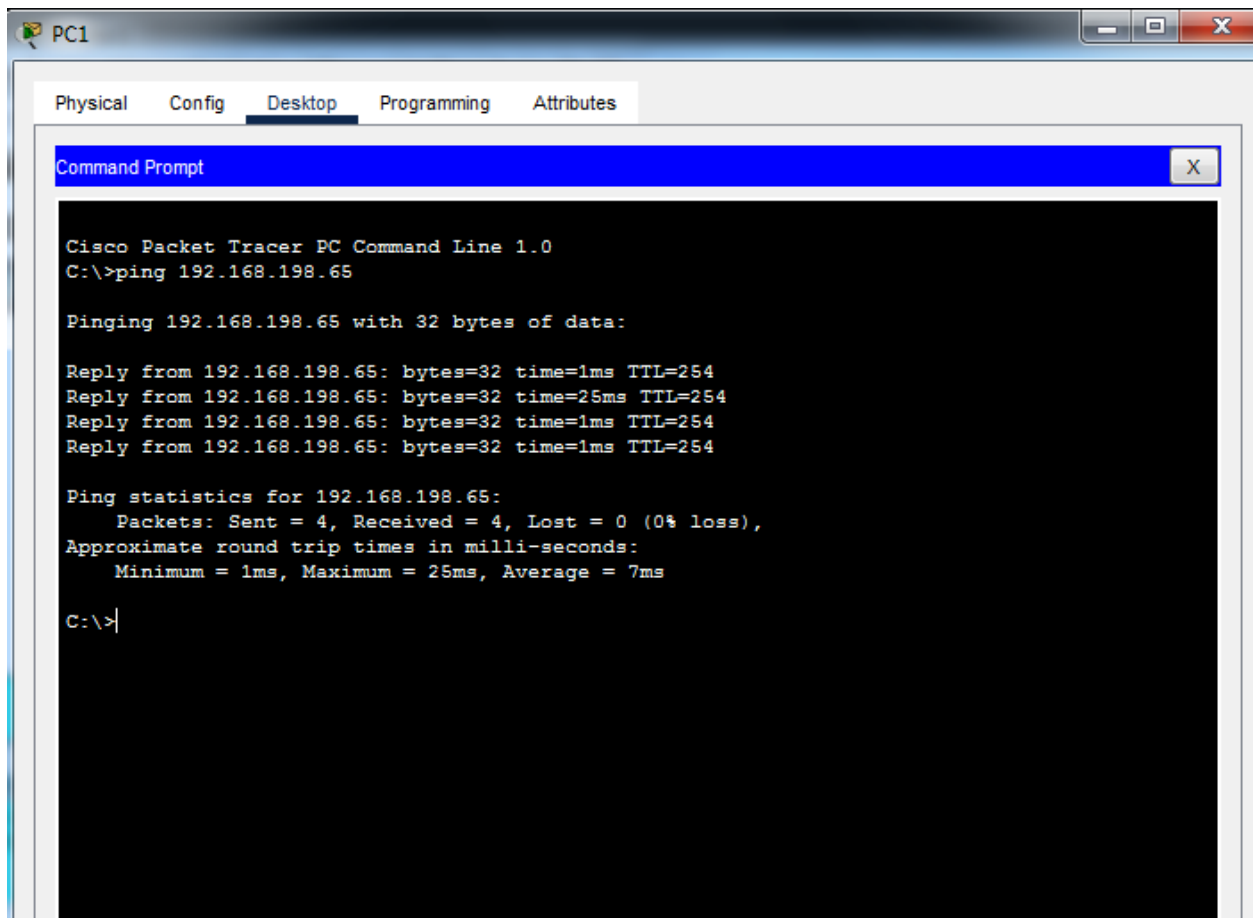
☐ Top

Final Topology in Packet Tracer



Ping Results

LAN1 to LAN 2



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC1. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and a Command Prompt window is open. The Command Prompt displays the output of a ping command from 192.168.198.65 to 192.168.198.65. The output shows four successful replies with varying round trip times (1ms, 25ms, 1ms, 1ms) and a TTL of 254. The ping statistics indicate that all four packets were sent and received with 0% loss. The approximate round trip times are: Minimum = 1ms, Maximum = 25ms, and Average = 7ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.198.65

Pinging 192.168.198.65 with 32 bytes of data:

Reply from 192.168.198.65: bytes=32 time=1ms TTL=254
Reply from 192.168.198.65: bytes=32 time=25ms TTL=254
Reply from 192.168.198.65: bytes=32 time=1ms TTL=254
Reply from 192.168.198.65: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.198.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 7ms

C:\>
```

LAN 1 to LAN 3

```
C:\>ping 192.168.198.129

Pinging 192.168.198.129 with 32 bytes of data:

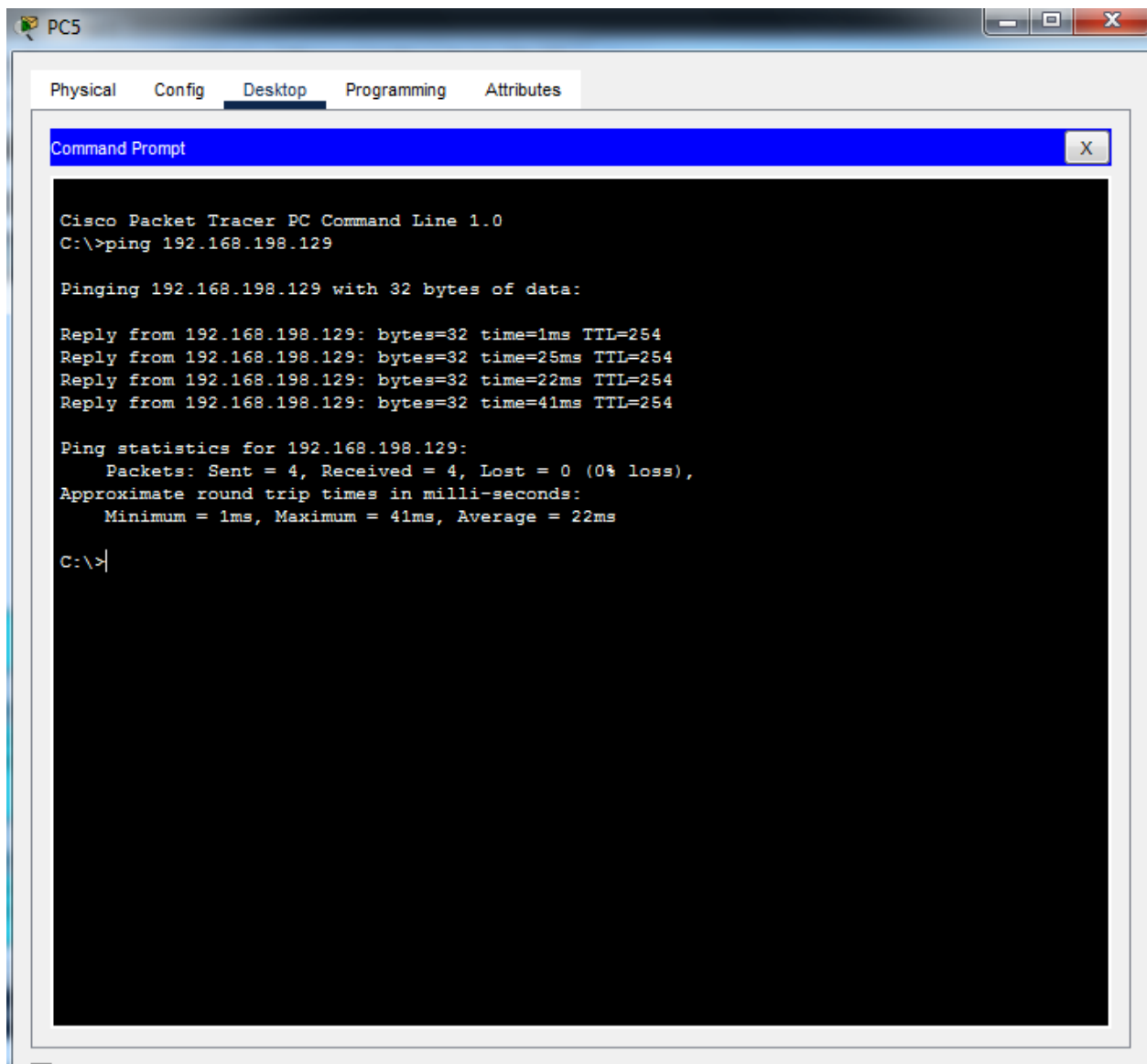
Reply from 192.168.198.129: bytes=32 time=26ms TTL=254
Reply from 192.168.198.129: bytes=32 time=2ms TTL=254
Reply from 192.168.198.129: bytes=32 time=21ms TTL=254
Reply from 192.168.198.129: bytes=32 time=2ms TTL=254

Ping statistics for 192.168.198.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 26ms, Average = 12ms

C:\>|
```

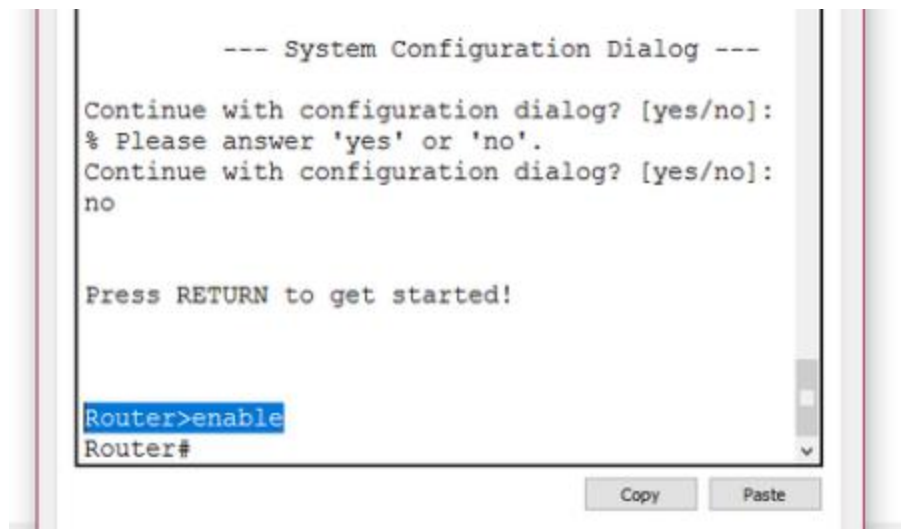
☐ Top

LAN 2 to LAN 3

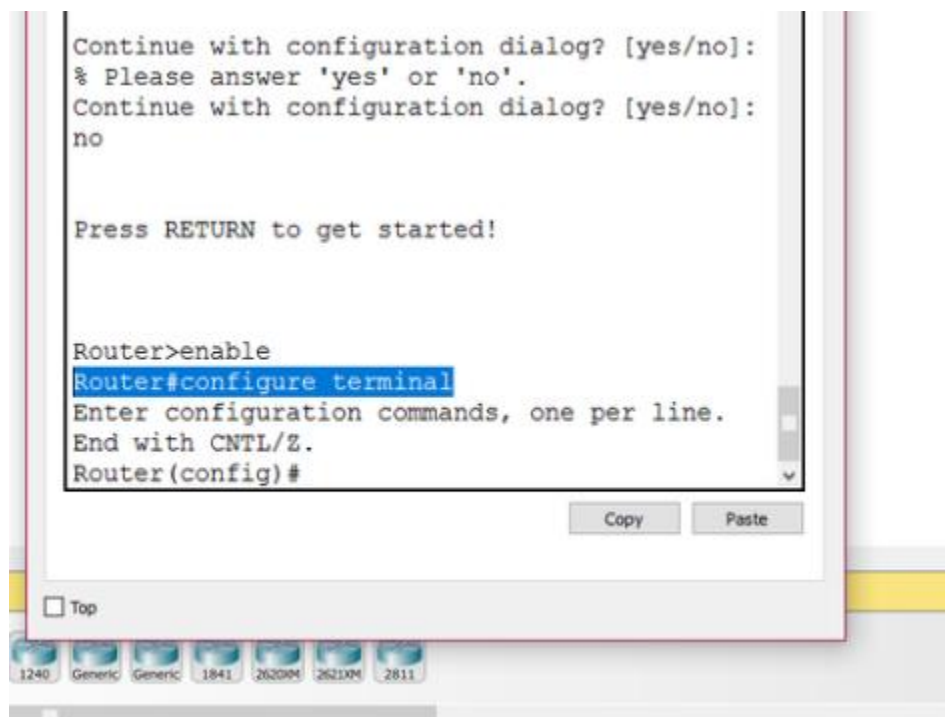


Password Implementation

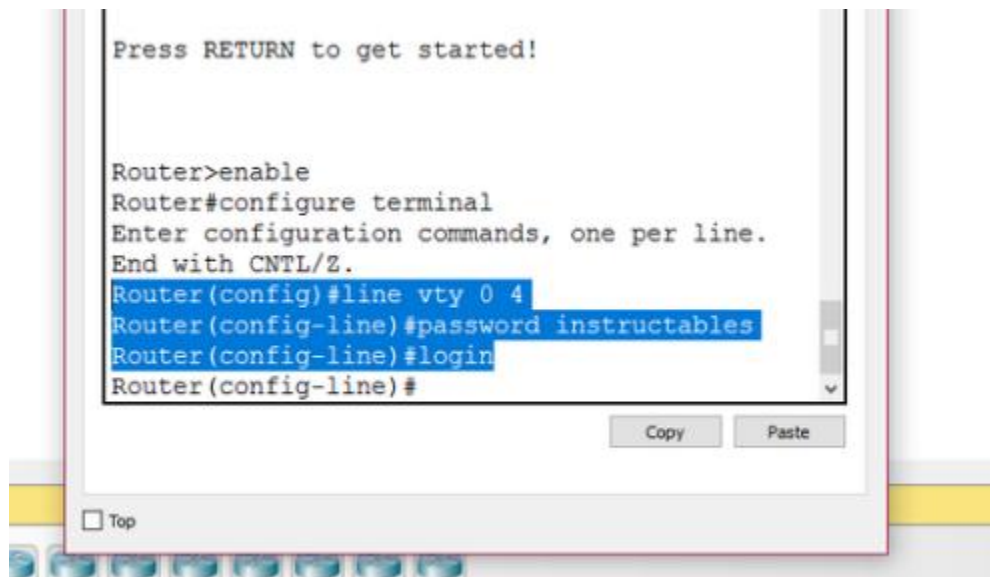
During the transition from privileged mode to general mode, the following instructions will be carried out:



In order to properly configure passwords, you need to enter the global configuration mode and use the instructions listed below:



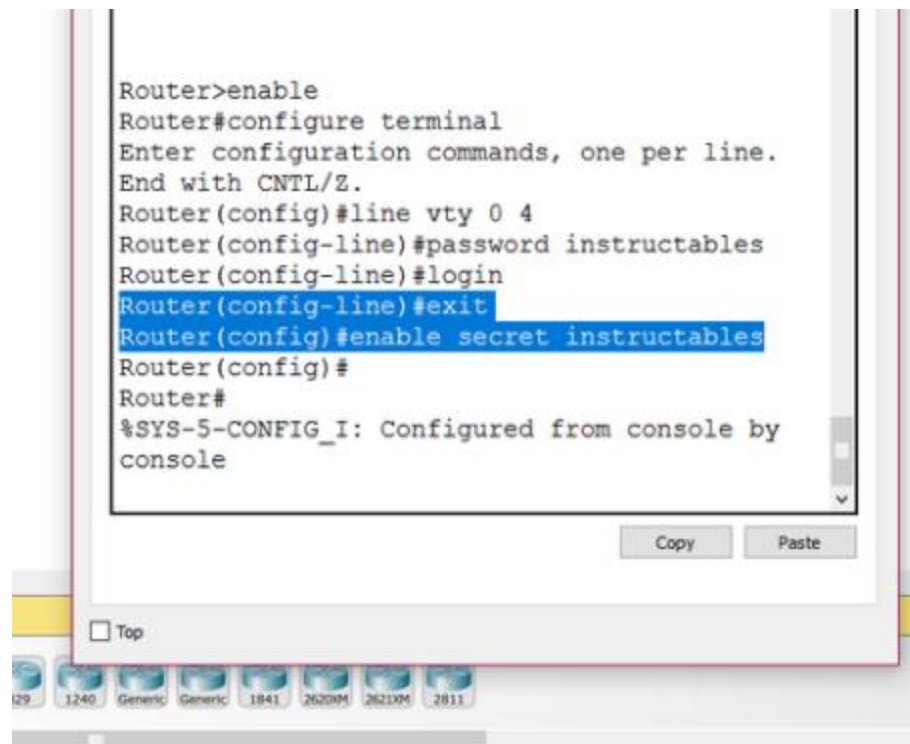
After the password has been generated, it must be made active inside the system's global configuration settings. To successfully finish the activation procedure, simply follow the detailed steps that are provided.

A screenshot of a Cisco Router terminal window. The window has a title bar with a 'Top' button. The terminal text shows the following sequence of commands and prompts: 'Press RETURN to get started!', 'Router>enable', 'Router#configure terminal', 'Enter configuration commands, one per line. End with CNTL/Z.', 'Router(config)#line vty 0 4', 'Router(config-line)#password instructables', 'Router(config-line)#login', and 'Router(config-line)#'. The last four lines of the terminal output are highlighted in blue. Below the terminal window, there are 'Copy' and 'Paste' buttons. At the bottom of the window, there is a taskbar with several icons.

```
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Router(config)#line vty 0 4
Router(config-line)#password instructables
Router(config-line)#login
Router(config-line)#
```

It is possible to discover, via repeated efforts to access each configuration mode, that the acquisition of privileged and global configuration modes requires the submission of a password. This may be determined by iterative attempts to access each configuration mode.



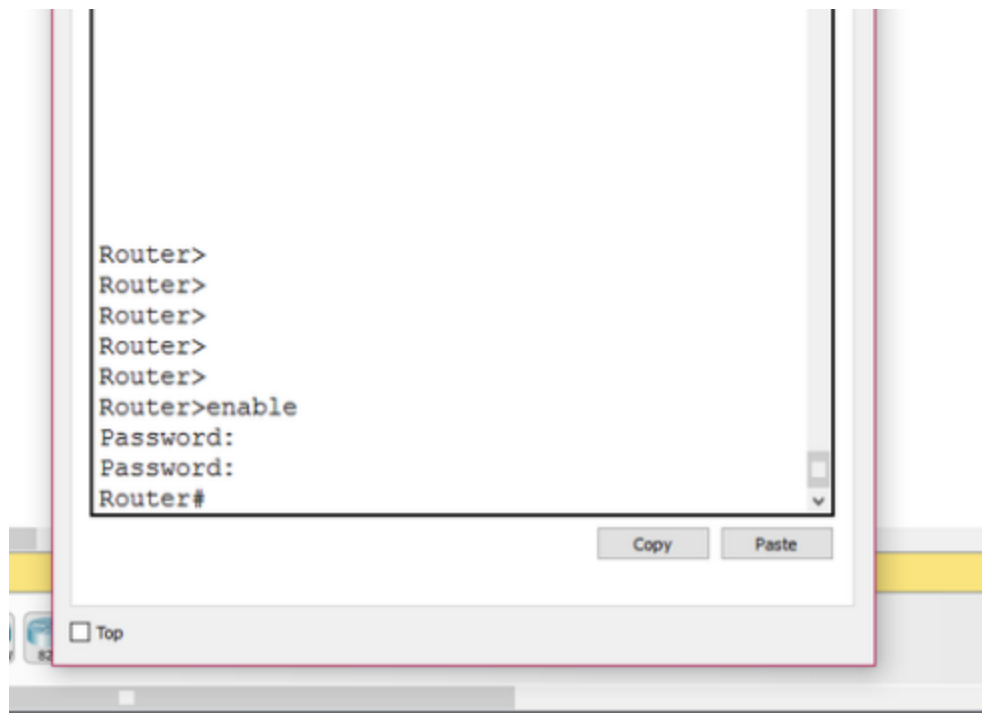
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Router(config)#line vty 0 4
Router(config-line)#password instructables
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret instructables
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by
console
```

Copy Paste

☐ Top

129 1240 Generic Generic 1841 26200M 26210M 2811

During this experiment, we will attempt to make a connection with the router by utilizing a console cable and serial communication on a laptop. This will allow us to get data from the router.



Explanation of Rip

When determining the best path, RIPv1 takes into account both the IP destination and the hop count. The original RIP protocol was improved with the introduction of RIPv2, which included the concepts of subnet masks and gateways. In addition, it is important to note that although RIPv1 broadcasts the routing table to all of the stations contained inside the linked network, RIPv2 makes use of a multicast address to accomplish the same goal. This measure is put into effect in order to lessen the strain that is placed on the network.

The Routing Information Protocol version 1 (RIPv1) is notable for the classful routing feature that it possesses. In the absence of subnet information and the capacity to perform VLSM, the periodic routing updates are of little utility. It is not feasible to have subnets of various sizes within the same network class because of this constraint. Within the framework of a network class, it is obligatory for all subnets to have dimensions that are equal to one another. Due to the fact that RIP does not have any router verification measures, it is susceptible to a wide variety of attacks.

According to the description provided in RFC 1723, the Routing Information Protocol version 2 (RIPv2) is a distance vector routing protocol that functions independently of the different types

of networks. The Border Gateway Protocol (BGP), which is a classless routing system, is distinguished by the fact that it transmits the subnet mask with network addresses in routing updates. This is one of the characteristics of the Border Gateway Protocol (BGP).

Tools to check connectivity

In order to enhance the visual and expeditious delivery of pings, one may opt to utilize the Add Simple PDU button.

Pings can be exchanged between any two IP-enabled devices that share at least one interface in common.

To initiate the transmission of a ping, please choose the Add Simple PDU button. Upon selection, the pointer will transform into a "packet" icon. Proceed by specifying the sending and receiving devices. The ping will then be dispatched. It is important to consider that pings need certain ports.

Upon receiving your request, the source device will place an ICMP or ARP packet (or both) into a designated buffer. These packets will remain in the buffer until you initiate the automated processes of capture, playback, or forwarding. The ping operation may be observed in real time by initiating packet transmission upon pressing one of these buttons.

The schedule

The Event List pane records the events that transpire throughout the dissemination of your Power Distribution Unit (PDU) over the network.

The Event List is designed to exclude any PDUs that have been subjected to filtering using the Event List Filters. The individuals in question have not been removed from the system; rather, you are now uninformed of their existence.

The unobservable nature of filtered events in the simulation results in enhanced computational efficiency, but with the caveat that all filtered protocol data units (PDUs) continue to exert an influence on the network.

Utilizing the Virtual Environment

The complete simulation may be executed automatically using the Auto Capture/Play feature.

By activating the Capture/Forward button, the packet will be transmitted to the next node in the network.

The findings of the study indicate

The temporal progression inside a Packet Tracer simulation does not adhere to a linear scale. The progression of time is dependent on the events that occur. Each instance of a Protocol Data Unit (PDU) being transmitted over a network is regarded as an occurrence or incident. The Event List is responsible for recording and organizing all instances of Protocol Data Unit (PDU) occurrences. It consists of many data fields that include:

The following is a compilation of many fields within the realm of events.

The visibility of an event is denoted by the existence of a "eye" indication within the field at the present moment of simulation. The symbol in question represents a packet that is now being shown inside the context of the scenario animation.

If the simulation scenario has been recently resumed, this variable will indicate the duration that has elapsed since the occurrence of the incident. Additionally, this serves as the reference point throughout the duration of the simulation.

The final recipient of the packet is denoted in this domain.

The current location of the packet is displayed in the "At Device" section.

The kind of the packet is shown in the "Type" column.